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1 Examination 1

UNIVERSITY EXAMINATIONS

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MAT1503

May/June 2020

Linear Algebra I

Duration : 2 Hours

100 Marks

EXAMINERS :

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SECOND : DR M DAVIES

Closed book and online examination, which you have to write within 2 hours and submit online, by no later than 17:00 on 25 June 2020.

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Save Frequently while working

Answer All Questions and Submit within the stipulated timeframe.

Late submission will not be accepted.

Part 1

Question 1 - [4 Marks]

Let C be an m by n matrix and k a scalar such that $kC = 0$. Which of the following statements is/are correct?

A. Either $k = 0$ or $C = 0$.

B. k is not zero and C is not the zero matrix.

☐ A.

Only A.

☐ B.

☐ C.

Both A and B.

☐ D.

Neither A nor B

☐ E.

Only B.

Question 2 - [4 Marks]

Let T be an m by n matrix.

Which of the following statements is/are correct?

A. $T = -T$ if and only if T is the m by n zero matrix.

B. $T = -T$ if T is an $m \times n$ symmetric matrix.

- ☐ A.
Neither A nor B.
- ☐ B.
Only B.
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 3 - [6 Marks]

Let A and B be 3×3 matrices. Given that

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 2 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Which of the following statements is/are correct?

$$\text{C. } (AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} \frac{8}{3} & -1 & -\frac{1}{3} \\ -2 & 1 & 0 \\ \frac{5}{3} & -1 & \frac{2}{3} \end{bmatrix}$$

$$\text{D. } (BA)^{-1} = A^{-1}B^{-1} = \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & 0 \\ -\frac{2}{3} & \frac{4}{3} & \frac{7}{3} \end{bmatrix}$$

- ☐ A.
Only C.
- ☐ B.
Neither C nor D.
- ☐ C.
Only D.
- ☐ D.
Both C and D.

Question 4 - [7 Marks]

Use Gaussian elimination to solve the following system of equations.

$$\begin{cases} 2x + 2y - 4z = 0 \\ 2x + 2y - z = 1 \\ 3x + 2y - 3z = 3 \end{cases}$$

Which of the following statements is/are correct?

A. $x = 8, y = -2, z = 1$ is the unique solution.

B. $x = -\frac{8}{3}, y = 2, z = \frac{1}{4}$ is the only solution.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Neither A nor B.
- ☐ C.
- ☐ Only B..
- ☐ D.
- ☐ Only A.

Question 5 - [5 Marks]

Let C and D be two square matrices of the same size. Let E be the inverse of D . If $\det(C) = 3$ and $\det(D) = 4$, what is the value of $\det(CE)$?

A. $\det(CE) = 12$.

B. $\det(CE) = \frac{3}{4}$.

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Neither A nor B
- ☐ D.
Both A and B.

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Question 6 - [4 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

What is the matrix form of the system, *i.e.*, $AX = B$?

C. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 9 \end{bmatrix}$

D. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

- ☐ A.
Neither C nor D.
- ☐ B.
Only D.
- ☐ C.
Only C.
- ☐ D.
Both C and D.

Question 7 - [6 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

Which of the following statements is/are correct?

C. The system has exactly one solution if

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

D. The system has exactly one solution if A is singular

- ☐ A.
- ☐ Both C and D.
- ☐ B.
- ☐ Only C.
- ☐ C.
- ☐ Only D.
- ☐ D.
- ☐ Neither C nor D.

Question 8 - [5 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

If A is the coefficient matrix of the above system; then using the inverse of given by

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Deduce which of the following statements is/are correct?

C. The solution of the system is $X = (0, -1, 2)$.

D. The solution of the system is given by $x = 1, y = 0, z = 2$

- ☐ A.
Only D.
- ☐ B.
Only C.
- ☐ C.
Neither C nor D.
- ☐ D.
Both C and D.

Question 9 - [5 Marks]

Given the following system of equations

$$\begin{cases} 2x + 3y - z = 1 \\ 4x + y + 2z = 5 \\ x + y + z = 2 \end{cases}$$

Find A_x , A_y , A_z from which calculate $\det(A_x)$, $\det(A_y)$, and $\det(A_z)$.

Which of the following statements is/are correct?

C. $\det(A_x) = -7$, $\det(A_y) = -3$ and $\det(A_z) = -12$.

Then $x = \frac{\det(A_x)}{\det(A)} = \frac{7}{11}$, $y = \frac{3}{11}$, $z = \frac{12}{11}$.

D. $\det(A_x) = -7$, $\det(A_y) = -3$ and $\det(A_z) = -12$.

Then $x = \frac{\det(A_y)}{\det(A)} = \frac{3}{11}$, $y = \frac{7}{11}$, $z = \frac{12}{11}$.

- ☐ A.
Only D.
- ☐ B.
Only C.
- ☐ C.
Both C and D.
- ☐ D.
Neither C nor D.

Question 10 - [4 Marks]

Let A be an n by n matrix. Suppose that in the system $A X = B$ all of the coefficients and constant terms are integers. Here B is a column vector.

Which of the following statements is/are correct?

C. If the solutions are all integers then $\det(A) = 1$.

D. The converse of the statement in C is true.

☐ A.
Neither C nor D.

☐ B.
Only D.

☐ C.
Only C.

☐ D.
Both C and D.

Question 11 - [6 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

What is the orthogonal complement of $\underline{v}$ onto $\underline{u}$?

A. $X = \left(\frac{2}{17}, 0, \frac{3}{17}\right)$.

B. $Y = \left(\frac{1}{17}, 0, \frac{3}{17}\right)$.

- ☐ A.
Neither A nor B.
- ☐ B.
Only B,
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 12 - [4 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

Which of the following statements is/are correct?

A. The area of the parallelogram bounded by $\underline{u}$ and $\underline{v}$ is $\sqrt{33}$ unit².

B. The area of the parallelogram bounded by $\underline{u}$ and $\underline{v}$ is $\sqrt{7}$ unit².

- ☐ A.
Neither A nor B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Only B.

Question 13 - [4 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

Find the perimeter of the parallelogram bounded by $\underline{u}$ and $\underline{v}$. Which of the following statement(s) is/are correct?

A. The perimeter of the parallelogram is *21 m*.

B. The perimeter of the parallelogram is *19 m*.

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 14 - [5 Marks]

Find an equation of the plane containing the point $(0, 1, 1)$ and perpendicular to the line passing through the points $(2, 1, 0)$ and $(1, -1, 0)$.

Which of the following statements is/are incorrect?

A. The equation of the plane is $x + 2y = 2$.

B. The equation of the plane is $-x - 2y = -2$.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Only B.
- ☐ C.
- ☐ Only A.
- ☐ D.
- ☐ Neither A nor B.

Question 15 - [4 Marks]

Find the parametric equations of the line that passes the point $(2, 3, 4)$ and is perpendicular to the plane $x + 2y - z = 0$.

Which of the following statements is/are correct?

A. If the direction vector is $\underline{v} = (-1, -2, 1)$ then the parametric equations of the line are $x - t = 2, y - 2t = 3, z + t = -4$.

B. If the line passes through the point $(2, 3, 4)$ then the parametric equations are

$$x - 2 = t,$$

$$y - 3 = 2t,$$

$$z - 4 = -t.$$

- ☐ A.
Only A.
- ☐ B.
Only B.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 16 - [3 Marks]

Solve for z given that $z + (1 - i) = 3 + 2i$.

Which of the following statements is/are correct?

A. $z = 2 + 3i$.

B. $z = 1 - i$ or $z = 3 + 2i$.

- ☐ A.
Only B.
- ☐ B.
Neither A nor B.
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 17 - [4 Marks]

Express

$$\frac{i}{i+1} \text{ in the form } a + ib, a, b \in \mathbb{R}$$

Which of the following answer(s) is/are the simplified correct answer(s) for the above form?

A. $i - 1$.

B. $\frac{1}{2} + \frac{1}{2}i$.

☐ A. B. $\frac{1}{2} + \frac{1}{2}i$.

☐ B. Both A and B.

☐ C. Neither A nor B.

☐ D. A. $i - 1$.

Question 18 - [8 Marks]

Use De Moivre's Theorem to express

$(\sqrt{3} + i)^8$ in the form $a + ib$, $a, b \in \mathbb{R}$

Which of the following statements is/are incorrect?

A. $\arg(\sqrt{3} + i)^8 = \frac{\pi}{6}$ and $(\sqrt{3} + i)^8 = 2^7 (1 + i)$.

B. $(\sqrt{3} + i)^8 = 2^8 \left(\frac{1}{2}i + \frac{\sqrt{3}}{2} \right)$.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Only B.
- ☐ C.
- ☐ Neither A nor B.
- ☐ D.
- ☐ Only A.

Question 19 - [4 Marks]

Find the cube roots of -1 in the form $a + ib$.

Which of the following answer(s) is/are the simplified correct answer(s) for the above form?

A.

$$z_0 = \cos \pi - i \sin \pi = -1 - 0i,$$

☐ $z_1 = \cos(2\pi) + i \sin(2\pi) = 1 + 0i,$

$$z_2 = 2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) = 1 + \sqrt{3}i.$$

B.

$$z_0 = 2(\cos \pi - i \sin \pi) = 2 + 0i,$$

☐ $z_1 = \cos \pi + 2i \sin \pi = -1 + 0i,$

$$z_2 = 3 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right) = 0 - 3i.$$

C.

$$z_0 = \cos \pi - i \sin \pi = -1 - 0i,$$

☐ $z_1 = \sqrt{2} \left[\cos \left(\frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{4} \right) \right] = 1 + i,$

$$z_2 = \sqrt{2} \left[\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right] = -1 + i.$$

D.

$$z_0 = \cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$$

○ $z_1 = \cos \pi + i \sin \pi = -1 + 0i,$

$$z_2 = \cos \left(\frac{5\pi}{3} \right) + i \sin \left(\frac{5\pi}{3} \right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Question 20 - [4 Marks]

Eliminate θ from the equations

$$4x = \cos(3\theta) + 3\cos\theta;$$

$$4y = 3\sin\theta - \sin(3\theta)$$

Which of the following statement(s) is /are incorrect?

A. $x^2 + y^2 = 1.$

B. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 2$

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 21 - [4 Marks]

Express $\cos(4\theta)$ and $\sin(5\theta)$ as polynomials in terms of powers of $\cos \theta$ and $\sin \theta$.

Which of the following statement(s) is/are correct?

- A. $\cos(4\theta) = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta,$
 $\sin(5\theta) = \sin^5 \theta - 10 \sin^3 \theta \cos^2 \theta + 5 \sin \theta \cos^3 \theta.$
- B. $\cos(4\theta) = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta,$
 $\sin(5\theta) = \sin^5 \theta - 15 \sin^3 \theta \cos^2 \theta + 5 \sin \theta \cos^3 \theta.$

- A.
- ☐ Only A.
- B.
- ☐ Only A.
- C.
- ☐ Both A and B.
- D.
- ☐ Neither A nor B.

Question 22 - [3 Marks]

Let

$$4x = \cos(3\theta) + 3\cos\theta$$

Express $\cos(4\theta)\sin(3\theta)$ in terms of multiple angles and in term of x .

Which of the following statement(s) is/are correct?

A. $\cos(4\theta)\sin(3\theta) = 2^7 x [\cos(4\theta) - 4\cos(2\theta) + 3]$.

B. $\cos(4\theta)\sin(3\theta) = 2^7 x^2 [\cos(4\theta) - 4\cos(2\theta) + 3]$.

- A.
- ☐ Only A.
- B.
- ☐ Neither A nor B.
- C.
- ☐ Both A and B.
- D.
- ☐ Only B.